

KS3 Character Representation

TOP SECRET

Initiating Codebreaker Protocol

How do computers represent text using binary?

Interview lesson | Chellaston Academy | Year 9 Computing

01000010

01001001

01001110

01000001

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Demonstrate

Reflect

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Mission Clearances Required

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Learning objectives moved forward so students know the mission early

By the end, you can...

- Explain why computers use binary.
- Decode 8-bit binary into text.
- Create a short binary message.

Success looks like...

- Use the reference table accurately.
- Keep each character as exactly 8 bits.
- Explain what a character set is.

MWB CHECK

One byte has how many bits?

- A: 4
- B: 8
- C: 16
- D: 32

Do Now Activity

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5 minutes | books open | answer in silence first

Thinking task

- Name communication systems that use only two states.
- Clue: one uses dots and dashes.
- Why can two states still carry a message?

Possible prompts

- on / off
- true / false
- open / closed
- dot / dash
- high / low voltage

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Binary Represents Data

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Video input, then MWB check

Key explanation

- Binary uses two symbols: 0 and 1.
- Computers use patterns of bits.
- Bits can represent text, images and sound.

[Open Nibblz binary video](#)

MWB CHECK

Which data can binary represent?

- A: images only
- B: text only
- C: text, images and sound
- D: numbers only

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How Binary Represents Text

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Video input, then a hinge check

Listening question

- What is the job of a character set?
- Why must everyone agree on the same codes?
- Write one key word you hear.

Open Nibblz text video

MWB CHECK

What does **01000001** mean in ASCII?

- A: A
- B: a
- C: !
- D: 65

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Codebook Rule

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Keep this short: agreed numbers become characters

The rule

- A character set is an agreed list of codes.
- ASCII gives characters a number.
- The computer stores the number as 8-bit binary.

A = 65

65 = 01000001

One code. One agreed meaning.

I Do: Decode BINARY

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Teacher models first; pupils follow on the HTML resource

01000010

B

01001001

I

01001110

N

01000001

A

01010010

R

01011001

Y

Decode one byte at a time using the reference table.

Open pupil HTML

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We Do: One Character

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Talk partner check before independent practice

Partner A

- Decode: 01000011
- Say which row you used.
- Explain why the prefix starts 010.

Partner B

- Check the answer and explanation.
- Point to the exact bit if you disagree.
- Then swap roles.

MWB CHECK

01000011 is...

A: A
B: C
C: c
D: B

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I Do: Encode Binary!

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Teacher models converting human text into 8-bit binary

Character	B	i	n	a	r	y	!
8-bit	01000010	01101001	01101110	01100001	01110010	01111001	00100001

Careful

- B and b are different ASCII codes.

Symbol

- The ! symbol begins 001.

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You Do: Create a Transmission

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Create binary only; your partner decrypts the message

Task

- Choose a secret message of up to 6 characters.
- Convert each character into 8-bit binary.
- Show your partner the binary only.

Example

- Message: Hello!
- Before sharing, cover or hide the character column.

[Open pupil HTML resource](#)

Binary code	Character
01001000	H
01100101	e
01101100	l
01101100	l
01101111	o
00100001	!

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ASCII And Unicode

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ASCII is useful, but it cannot represent every language well

ASCII

- Older character set.
- Common English letters, digits and punctuation.
- Useful for simple messages.

Unicode

- Much larger character set.
- Represents many languages.
- Also includes many symbols.

Video summary

Listen for: what ASCII stands for, what a character set is, and how many characters Unicode can currently store.

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Quick Check: Character Set

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Multiple choice

What is a character set?

- A** An agreed list of codes for characters.
- B** A type of computer screen.
- C** A password with letters and numbers.
- D** A program that changes files into images.

Quick Check: ASCII

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Multiple choice

What does ASCII stand for?

- A** American Standard Code for Information Interchange.
- B** Automatic Symbol Code for Internet Instructions.
- C** Advanced System Code for Input Interpretation.
- D** Alphabetic Standard Code for Internal Information.

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Quick Check: Unicode

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Multiple choice

Unicode can represent...

- A** Fewer than 128 characters.
- B** The same characters as ASCII only.
- C** Over a million possible code points.
- D** Only capital letters and digits.

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Stretch: Decode A Sentence

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Remember that a space is a character too

Decode this sentence. Use ASCII 00100000 for a space.

01000011	01001111	01000100	01000101
00100000	01010010	01000101	01000100
C	O	D	E
space	R	E	D

Stretch prompt

- Why does the computer need a binary code for a space?

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